CRISPR-CAS9 GENE EDITING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on CRISPR-Cas9 Gene Editing

Question 1

10.0 points

Examine the diagram below that illustrates the Cas9 protein–sgRNA complex docking onto target DNA. Describe the molecular interactions between the sgRNA and the target DNA that are critical for specificity. Identify any key residues or structures in the Cas9 protein involved.

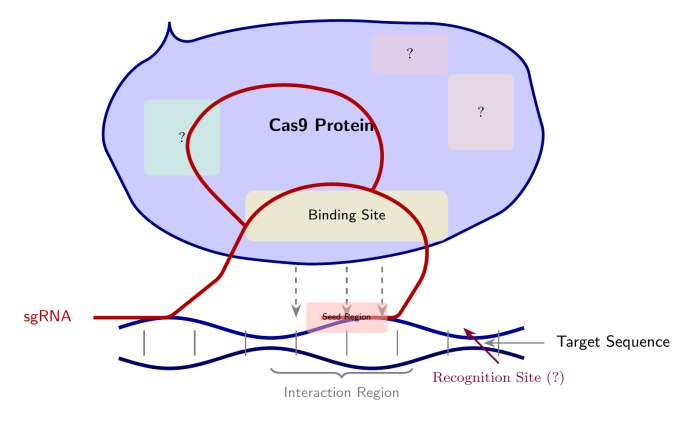

Fig. 1. Diagram of the Cas9 protein-sgRNA complex docking onto target DNA. The diagram should show the Cas9 protein with key domains labeled, including the RuvC, HNH, and PAM-interacting domains. The sgRNA should be depicted as a guide RNA strand bound to the target DNA, with the seed sequence highlighted. The DNA should be shown as a double helix with the target sequence complementary to the sgRNA. Highlight key residues or structures in the Cas9 protein that are involved in specificity, such as those interacting with the PAM site and the seed sequence of the sgRNA. Use arrows or labels to indicate critical molecular interactions between the sgRNA and the target DNA, emphasizing specificity-determining regions. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • DNA • RNA • PAM • RuvC • HNH • Watson • Crick • base editing window • critical threshold The diagram will FAIL review if any of these appear.

Question 2

10.0 points

Examine the diagram below illustrating the PAM sequence recognition and DNA unwinding mechanism in CRISPR-Cas9. Discuss the role of the PAM sequence in Cas9 activation and how DNA unwinding is achieved.

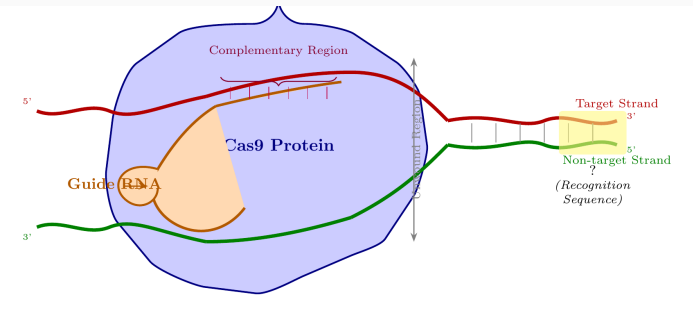

Fig. 2. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: PAM, DNA; The diagram includes the label "PAM Recognition", which contains the forbidden word "PAM" and directly reveals part of the answer the student is supposed to discuss.; The diagram contains labels that directly state the processes the student is asked to explain, specifically "Activation" and "Strand separation".. A detailed schematic of the CRISPR-Cas9 mechanism. The diagram must show the Cas

Question 3

10.0 points

Examine the diagram below illustrating DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). Compare these pathways in terms of accuracy and potential applications in gene editing.

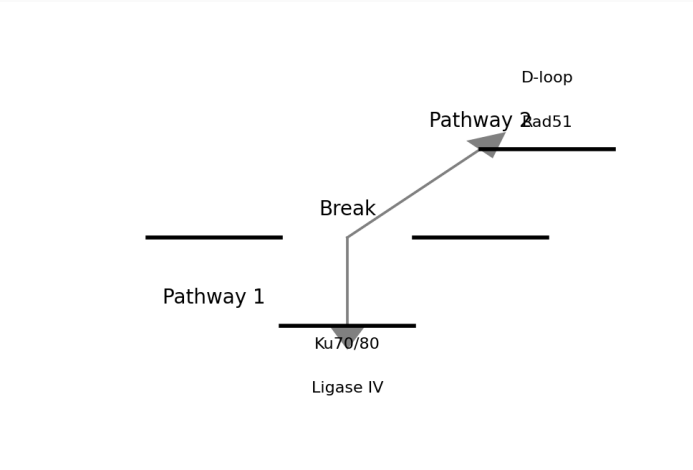

Fig. 3. A diagram illustrating DNA double-strand break repair pathways, focusing on non-homologous end joining (NHEJ) and homology-directed repair (HDR). The diagram should depict a DNA double-strand break, with two separate pathways branching from it. The NHEJ pathway should show direct ligation of the broken ends, possibly with some loss of nucleotides, and involve proteins like Ku70/80 and DNA ligase IV. The HDR pathway should show the use of a homologous template for accurate repair, involving proteins like Rad51 and the formation of a D-loop. Label each pathway clearly and indicate the key proteins involved in each process. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • DNA • NHEJ • HDR • base editing window • syndrome table • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 4 10.0 points

Analyze the mismatch tolerance in the diagram below for off-target effect prediction in CRISPR-Cas9 gene editing. How do different types of mismatches affect the likelihood of off-target effects?

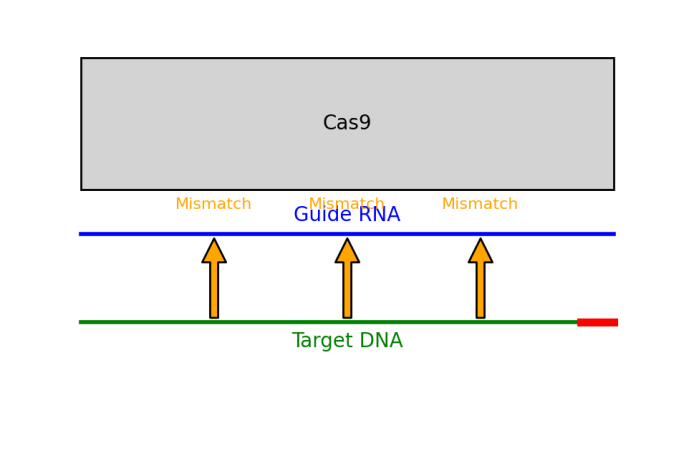

Fig. 4. A CRISPR-Cas9 complex diagram showing the Cas9 protein, guide RNA, and target DNA. Highlight potential mismatch sites between the guide RNA and the target DNA. Label the guide RNA, target DNA, PAM site, and mismatch sites. Use arrows to indicate where mismatches can occur and how they might affect the binding of the guide RNA to the target DNA. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • PAM • However • base editing window • infeasible region • feasible region • threshold region The diagram will FAIL review if any of these appear.

Question 5 10.0 points

Examine the base editing window in the diagram below for cytosine base editors (CBEs). Describe the CBE's base editing window and discuss the implications for specificity and efficiency in gene editing projects.

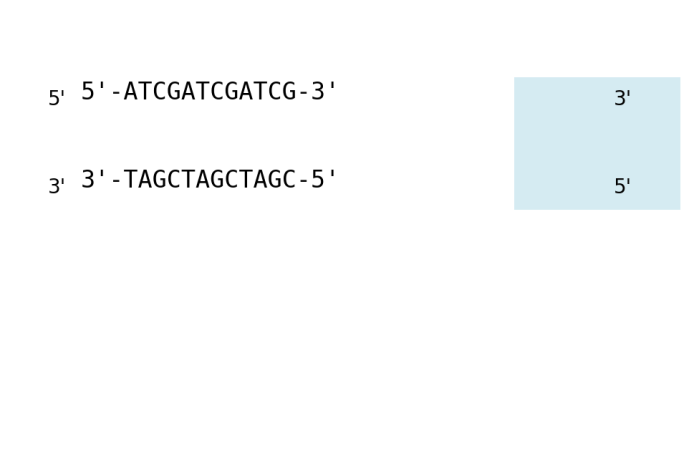

Fig. 5. A DNA sequence with a highlighted base editing window for cytosine base editors (CBEs). The diagram should show a double-stranded DNA helix with labeled nucleotide bases (A, T, C, G). Highlight the specific region where cytosine bases are targeted for editing, indicating the editing window. Include labels for the 5' and 3' ends of the DNA strands, and clearly mark the editing window with a different color or shading to distinguish it from the rest of the sequence. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • CBE • RNA • Increased • base editing window • infeasible region • feasible region • threshold region The diagram will FAIL review if any of these appear.